

Introduction to Exploratory Data Analysis

Course 2. Data sanity

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Typical scenario in data analysis/prediction/inference

- Large datasets

- Data from several sources / sensors

 - Requires synchronization

 - Different sampling frequencies

- Possibly collected by someone else

 - No control or info on acquisition quality

 - Possibly preprocessed, filtered

Data sanity

Data processing involves a set of decisions **D#**, bigger or smaller, which affect

- Performance of next analysis steps

- The extent to which data reflects reality

Data Sanity

- Clean data, as close as possible to the ideal expected by theory

- Unaltered data, as close as possible to the reality that produced them

Case study

Wearables, Samsung Gear S3 Frontier smartwatch
Digital phenotyping

One subject, $\approx$ 3 months of recordings

Date	File	Dimension	Sampling frequency
Accelerometer	linacc.csv	5.46 GB	NA (50 Hz Samsung)
Gyroscope	gyr.csv	4.42 GB	NA (50 Hz Samsung)
Heart rate	hrm.csv	888 MB	5 Hz
Sleep	sleep.csv	8 KB	NA
Steps	step.csv	652 KB	NA

Data source: <https://robotics.ntua.gr/icassp2024-e prevention-spgc/>

Accelerometer

Import .csv as Pandas DataFrame: `linacc = pandas.read_csv("linacc.csv")`

Number of samples: `linacc.shape[0]` = 102964745

Preview the pandas frame: `linacc.head(8)`

Var1	X	Y	Z	time	day_index
0	0.14	0.07	0.07	00:00:00.003241	0
1	0.07	0.14	0.07	00:00:00.053274	0
2	0.14	0.14	0.07	00:00:00.103308	0
3	-0.14	-0.07	0.07	00:00:00.153342	0
4	0.21	0.14	0.14	00:00:00.203376	0
5	0.21	0.14	0.14	00:00:00.253409	0
6	-0.14	0.07	-0.07	00:00:00.303442	0
7	0.07	0.07	0.14	00:00:00.353476	0

Accelerometer

Getting the sample frequency

We cannot assume continuous acquisition throughout the entire period.

Most frequent value of the time period between to consecutive samples:

```
scipy.stats.mode(np.diff(linacc['time'])): 00:00:00.050032
```

Largest value of the time period between two consecutive samples:

```
numpy.max(diff(linacc['time'])): 17:01:27.193956
```

$$F_s \approx 20\text{Hz}$$

Gyroscope

number of samples: `gyr.shape[0]` = 103528025

Var1	X	Y	Z	time	day_index
----	-----	-----	-----	-----	-----
0	0.027	-0.019	-0.083	00:00:00.003241	0
1	-0.001	-0.015	-0.086	00:00:00.053274	0
2	0.063	-0.034	-0.083	00:00:00.103308	0
3	-0.009	-0.021	-0.042	00:00:00.153342	0
4	0.048	-0.018	-0.033	00:00:00.203376	0
5	0.037	0.031	-0.111	00:00:00.253409	0
6	0.020	-0.023	-0.086	00:00:00.303442	0
7	0.027	-0.020	-0.073	00:00:00.353476	0

Giroscop

Most frequent value of the time period between to consecutive samples:

```
scipy.stats.mode(np.diff(gyr['time'])): 00:00:00.050032
```

Largest value of the time period between two consecutive samples:

```
numpy.max(diff(gyr['time'])): 16:20:10.137379
```

$$F_s \approx 20\text{Hz}$$

To check: `gyr['day_index'].iloc[-1] == linacc['day_index'].iloc[-1]`

Heart rate

Number of samples: `hrm.shape[0]` = 25880182

Var1	heartRate	rRInterval	time	day_index
0	62	1025	00:00:00.165496	0
1	62	1025	00:00:00.364978	0
2	62	1025	00:00:00.565292	0
3	62	1025	00:00:00.765829	0
4	62	1025	00:00:00.965379	0
5	62	1012	00:00:01.165512	0
6	62	1012	00:00:01.365645	0
7	62	1012	00:00:01.565809	0

Data dimension

D#1. Do we need all this data?

Depending on the purpose of the research, the sampling frequency might be too low or too high.

Do we need to synchronize with data from other sources, having different F_s ?

We do not need all the data.

Arguments?

Data dimension

D#2. How should we handle size reduction?

By eliminating sampling (decimation, downsampling)?

By retaining some features/characteristics of the eliminated samples (e.g. short-time analysis)?

Short-time analysis, because the research involves the characterization of the physical activity of the subject and therefore will require feature engineering.

Short-time analysis

Not the subject of today's lecture, but

Involves segmenting the signal into windows of predefined length and analyzing the windows / computer characteristics such as

1. Acc/gyro

- 1 Statistics: min, max, mean, variance
- 2 "Magnitude" (norm), Energy
- 3 Number of / frequency of spikes

2. HR

- 1 Statistics
- 2 Spectral characteristics (ideally on frequency bands)
- 3 Entropy

Data dimension

D#3. What is a appropriate downsampling factor / window size in short-time analysis?

Compromise between computation time and relevance.
Implies knowledge on the measured phenomena.

Literature says a 5 minutes window is ok.

Missing data

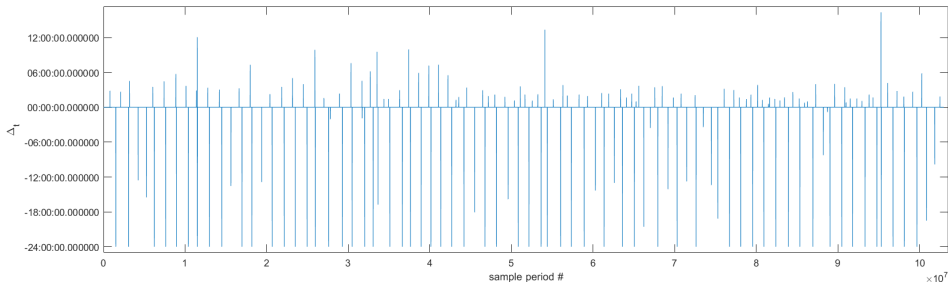
```
N = linacc['day_index'].iloc[-1] * 24 * 3600 * 20 = 152064000
```

```
Nreal = linacc.shape[0] = 103528025
```

32.29% missing data in ACC.

31.92% missing data in GYRO.

31.92% missing data in HR.



Missing data

D#4. What to do about missing data?

In this case, nothing: they are too many to be estimated and are not isolated.

However, ACC, GYR and HR are measured with the same device.
The fact that from GYR and HR the same percentage of data is missing, suggests that the device was not worn during those times.
the additional 0.37% missing in ACC might be a sensor problem.

We can look for the time samples where there is GYR data but no ACC and, if they are isolated, we can estimate ACC (e.g. with interpolation methods).

Missing data

D#5. What about the case of short-time analysis?

Discard the windows in which $x\%$ of data is missing and interpolate the remaining missing data.

Data quality

```
linacc['X'] [min, max] = [-86.7168, 44563]  
linacc['Y'] [min, max] = [-88.1595, 88.1765]  
linacc['Z'] [min, max] = [-87.8484, 87.6453]
```

Human movement range: [-19.6, 19.6]

```
gyr['X'] [min, max] = [-2294, 2292]  
gyr['Y'] [min, max] = [-2294, 1842]  
gyr['Z'] [min, max] = [-1333, 2264]
```

Human movement range: [-573, 573]

```
hrm['heartRate'] [min, max] = [-10, 220]
```

Heartbeat range: [0, 255]

```
hrm['rRInterval'] [min, max] = [-31603, 28669]
```

RR interval in humans: [0, 2000]

Data quality

D#6. What to do about the wrong data?

First, find out how many they are.

```
linacc.loc[(linacc['X']<-19.6 | linacc['X']>19.6, 'X')] = numpy.nan
```

```
linacc['X'].isna().sum() = 26329
```

```
linacc.isna().T.any().sum() = 75134 (0.7%)
```

```
gyr.isna().T.any().sum() = 60526 (0.6%)
```

```
hrm.isna().T.any().sum() = 400600 (1.6%)
```

Interpolation. What method? Linear, polynomial? Band pass filter?
Depends on data type and number of consecutive samples missing.

Data synchronization

```
linacc.shape[0] = 102964745  
gyr.shape[0] = 103528025
```

We have samples from one sensor, but not for the other(s).

D#7. Again, how to treat missing data?

Sleep data

Number of samples: `sleep.shape[0]` = 109

Var1	start_time	end_time	start_date_index	end_date_index
0	15:28:33.063159	15:47:33.824178	-1	-1
1	23:27:52.337799	08:02:13.080403	-1	0
2	22:54:38.945568	07:49:26.605778	0	1
3	22:43:54.632989	23:18:14.918784	1	1
4	23:41:15.843560	23:58:05.324129	1	1
5	20:47:18.743371	20:55:51.786747	2	2
6	12:59:52.397089	13:02:52.518478	3	3
7	22:16:33.760226	07:11:55.339440	4	5

Data relevance

D#8. Can we compare days that have 24h recordings to days that have 4h of recordings?*

Remove sleep data for days with less than x hours..

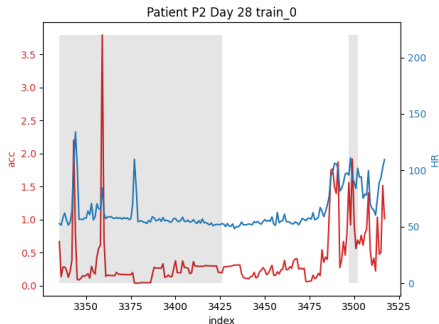
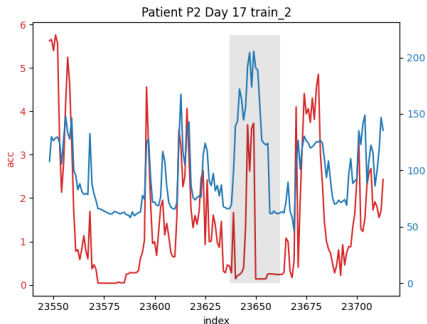
*This question must be asked in context: how will the data be used later? How will they be interpreted? What algorithm will process them, with what purpose and how does it work?

Data quality

D#9. Is data trustworthy?

Sleep periods are in gray. Can a person sleep with 200 bpm?

Should sleep data be re-estimated altogether? Should it be cross-checked/validated with ACC/HR?



Steps

Number of samples: `step.shape[0]` = 13585

Var1	total Steps	steps Walking	steps Running	distance	calories	start_time	end_time	start_date _index	end_date _index
0	17	17	0	12.90	0.58	07:55:00	08:06:07	0	0
1	111	111	0	71.62	3.61	08:06:07	08:08:33	0	0
2	90	90	0	58.99	2.9	08:08:33	08:09:01	0	0
3	21	21	0	12.06	0.69	08:15:00	08:26:00	0	0
4	34	34	0	21.31	1.08	08:38:37	08:42:44	0	0
5	101	101	0	72.08	3.43	08:42:44	08:43:00	0	0
6	97	97	0	73.94	3.43	08:43:00	08:44:00	0	0
7	14	14	0	11.17	0.46	08:45:16	08:48:43	0	0

Same analysis as above.

Further reading

Here you can find a good list of issues related to data cleaning and indications on how to solve them.